

Project 3 ML4T

Longtai Liao

lliao32@gatech.edu

Abstract:

This project aims to introduce and explore the usage and performance of four CART regression algorithms: Decision Tree Learner, Random Tree Learner, Bag Learner, and Insane Learner which is combined with Linear Regression and Bagging.

Introduction:

The performance of the above ML models is analyzed by comparison between the in-sample data set and the out-of-sample data set. Overfitting and underfitting are one of the topics we will explore. Multiple measure metrics such as RMSE, variance, and MAE (Mean Absolute Error)

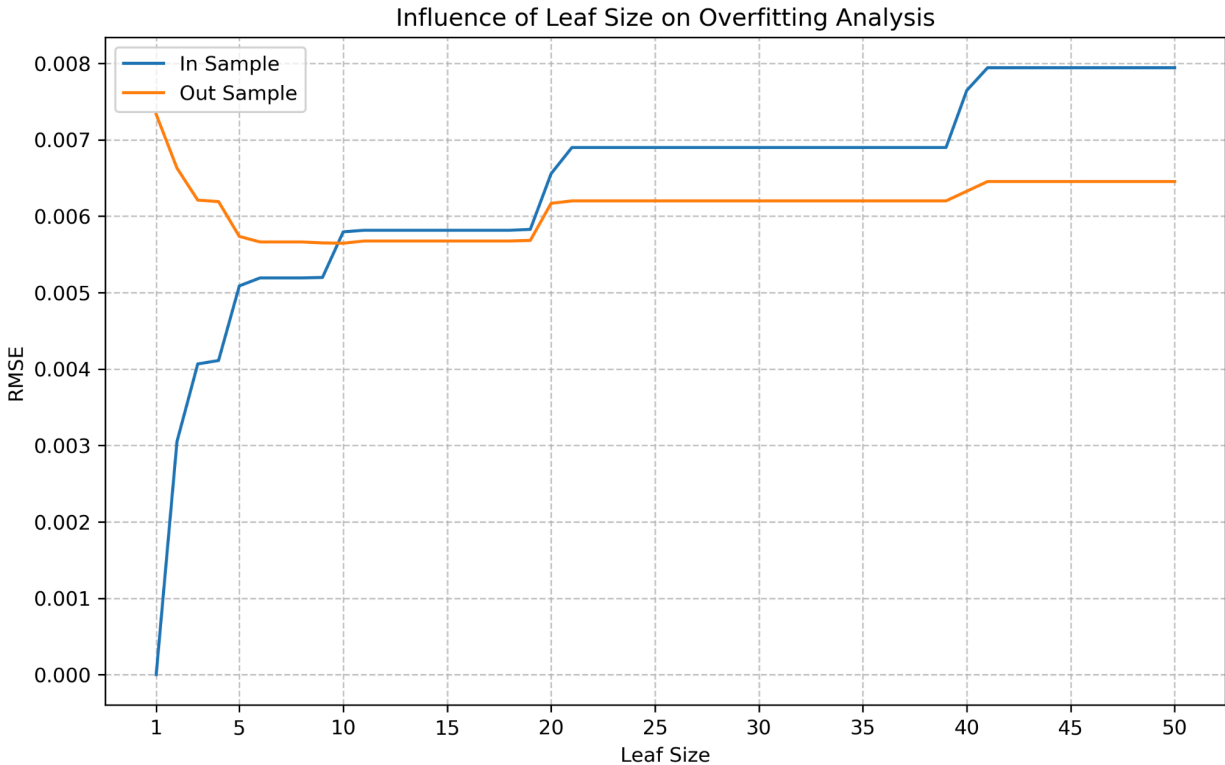
Methods:

In this project, we are running in an ML4T environment and the training models are constructed as instances in files DTLearner.py, RTLearner.py, BagLearner.py, and insaneLearner.py. We will use Istanbul.csv as the training set and generate corresponding charts via testlearner.py

Discussion:

Experiment 1:

The result of experiment 1 indicates that leaf size influences the overfitting situation. Figure 1 shows that when the leaf size is relevantly small (between 1 and 10), the out-sample RMSE is larger than the in-sample RMSE. Overfitting occurs during this range, and as the leaf size increases, it declines. When leaf size is 10, in the sample and out sample RMSE almost reached the same value, this is when overfitting occurs. As the leaf size keeps increasing, RMSE on both sides increases and in-sample RMSE becomes larger, which means the prediction is getting less accurate.



Figure_1

Experiment 2:

Experiment 2 analyzes the effect of bagging on overfitting. In this experiment, I am using 20 bags for the DTLearners set, and Figure 2 reveals the relationship between overfitting and leaf size of the DTLearner inside. Overfitting still exists when leaf size is less than 10. Generally speaking, compared with experiment 1- Figure 1, the overfitting is significantly reduced. Out of sample RMSE in bagging always stays around 0.0047 when leaf size is below 40 and will reach 0.005 until 40 leaf size. In comparison, out of the sample RMSE is always above 0.006 and the highest will even reach 0.008 for DTlearner in experiment 1.

Although bagging can greatly reduce overfitting, it cannot eliminate it. Bagging might slightly increase in-sample error while improving out-of-sample performance. However, it will produce more consistent performance between training and testing.

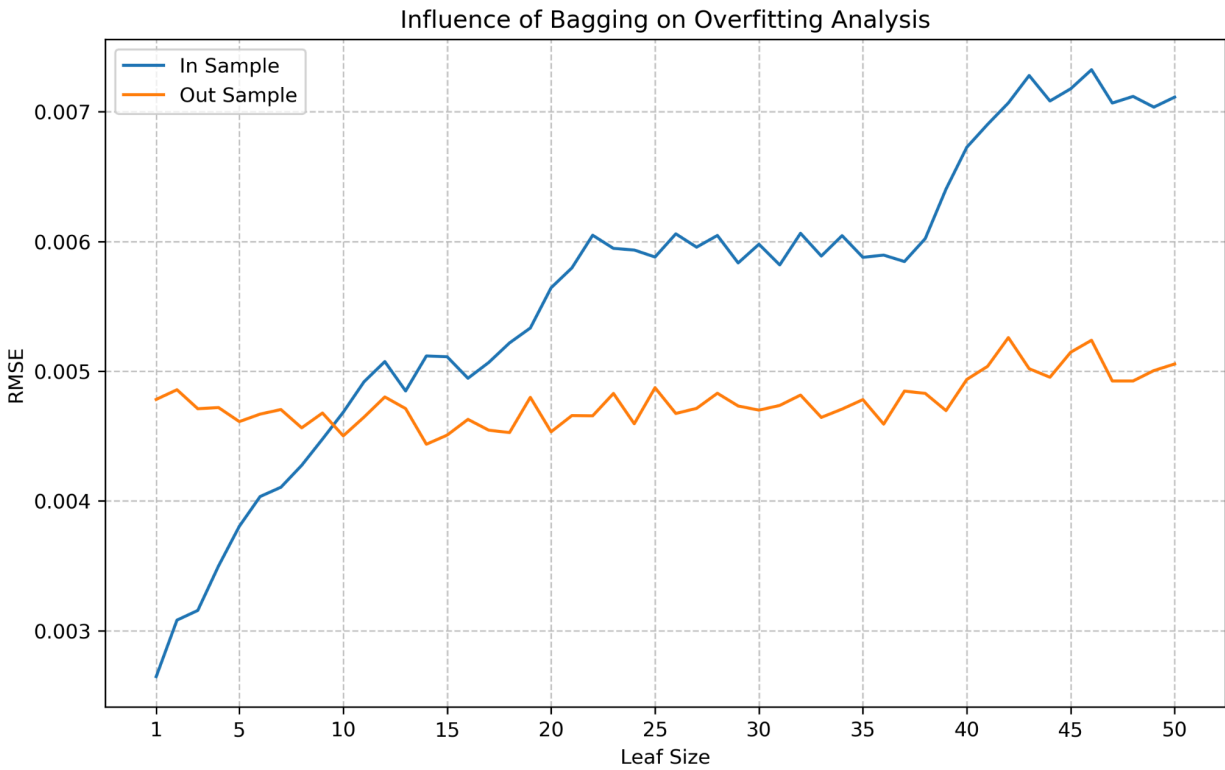


Figure 2

Experiment 3:

In experiment 3. Two new quantitative measures are introduced to analyze the performance of DTLearner and RTLearner. The two metrics are Median Absolute Error Analysis(MedAE) and Training time Analysis.

Figure 3 analyzes the MedAE of DTLearner and RTLearner in terms of the changes of leaf size. As the experiment figure 3 shows, DTLearner has more stable performance across different leaf sizes while RTLearner shows more variability. RTLearner performs better when the leaf size is small (between 5 to 18). From this experiment, we can conclude that DTLearner is better for accuracy as it has a lower and stabler MedAE.

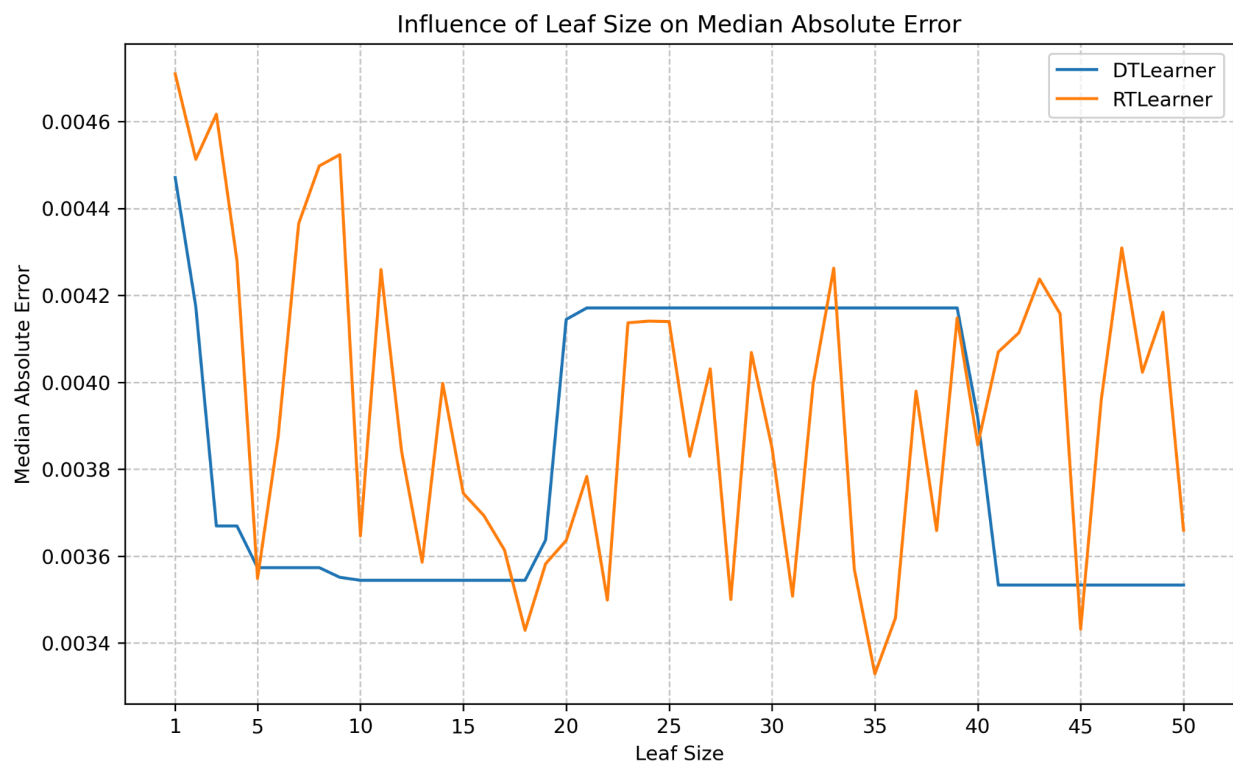


Figure 3

Figure 4 analyzes the training time performance in terms of the Data size. Both DTLearner and RTLearner set their leaf size as 10, while 10 is the leaf size when they perform best in previous experiments. From this experiment, we can tell that RTLearner is consistently faster in training across all data sizes, and DTLearner's training time increases more dramatically when data size increments. The difference gap between DTLearner and RTLearner also enlarged as data size increased. In the whole test range, RTLearner keeps a relatively stabler and lower training time. To conclude, RTLearner is much more efficient in terms of computation as it has faster training times.

Combining MedAE and Training Time experiment, we can find that DTLearner has better predictive performance but has a higher computational overhead. RTLearner sacrifices its accuracy for significant efficiency in training time. So none of those learners is the best, we should make the right choice based on the specific use case.

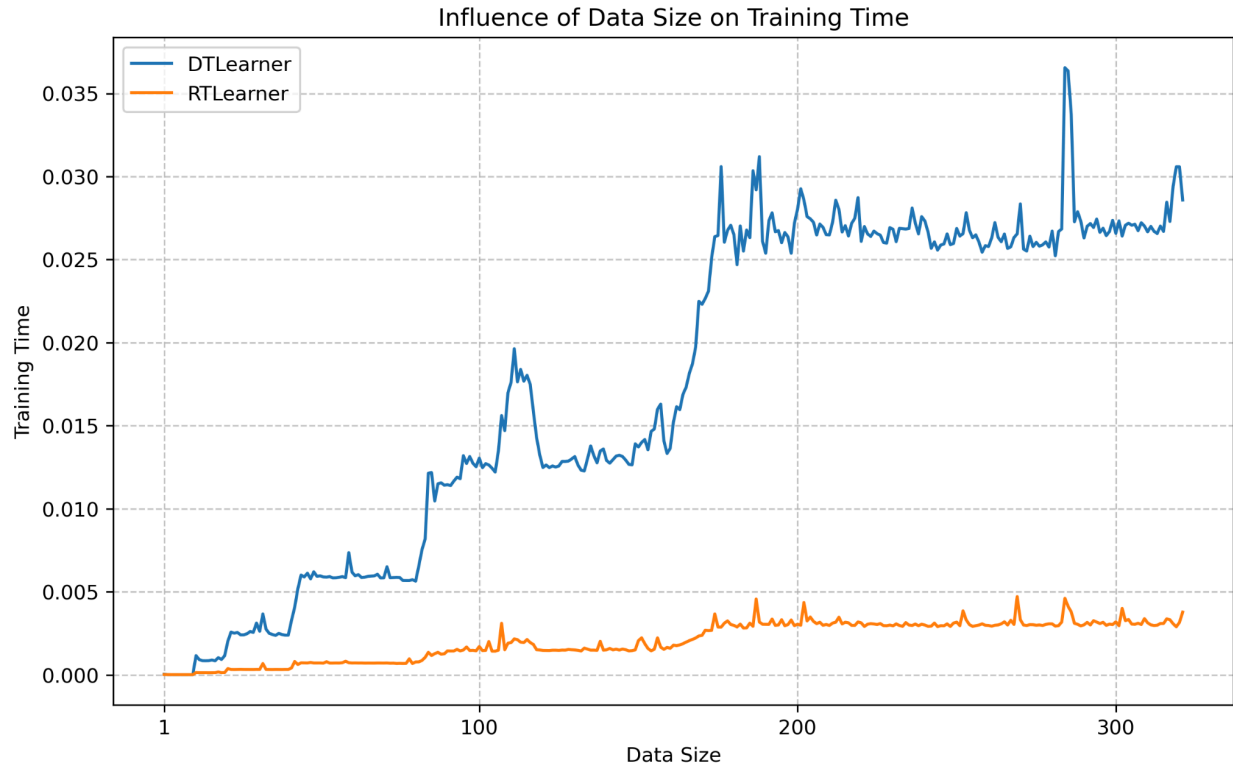


Figure 4

Summary:

In summary, for experiment 1, we can conclude that a small leaf size (1-10) leading to overfitting and a leaf size around 10 is the optimal RMSE performance. Overall prediction accuracy will reduce in larger leaf size. we can find from experiment 2 that the bagging strategy will significantly reduce the overfitting effect compare with DTLearner and keep a stable RMSE in around 0.0047. Experiment 3 compared with DTLearner and RTLearner in efficiency and accuracy. DTLearner shows more stable and generally better accuracy while RTLearner has better computational efficiency. For future investigation, we might explore the hybrid approaches to combine the strengths of DT and RT learners.